

String vs StringBuilder vs StringBuffer

Java Methods Comparison and Big Picture

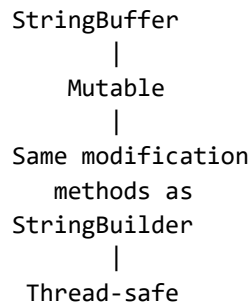
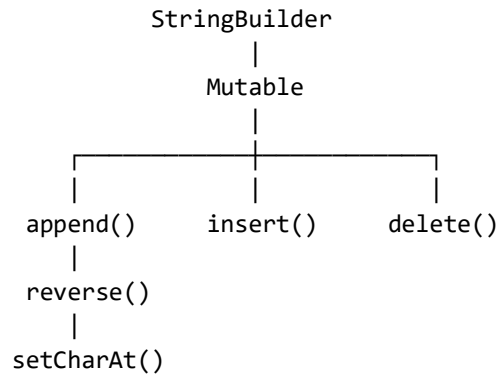
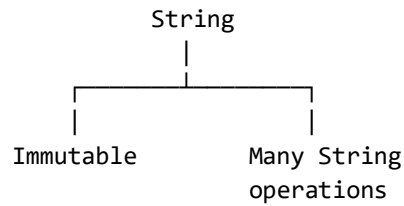
16. The Most Useful Comparison

The following table compares commonly used methods of String, StringBuilder, and StringBuffer. A check mark means the method is available directly in that class.

Method	String	StringBuilder	StringBuffer
length()	✓	✓	✓
charAt()	✓	✓	✓
substring()	✓	✓	✓
subSequence()	✓	✓	✓
indexOf()	✓	✓	✓
lastIndexOf()	✓	✓	✓
getChars()	✓	✓	✓
codePointAt()	✓	✓	✓
codePointBefore()	✓	✓	✓
codePointCount()	✓	✓	✓
offsetByCodePoints()	✓	✓	✓
chars()	✓	✓	✓
codePoints()	✓	✓	✓
toString()	✓	✓	✓
append()	✗	✓	✓
insert()	✗	✓	✓
delete()	✗	✓	✓
deleteCharAt()	✗	✓	✓
reverse()	✗	✓	✓
setCharAt()	✗	✓	✓
setLength()	✗	✓	✓
capacity()	✗	✓	✓
ensureCapacity()	✗	✓	✓
trimToSize()	✗	✓	✓
contains()	✓	✗	✗
startsWith()	✓	✗	✗
endsWith()	✓	✗	✗
equalsIgnoreCase()	✓	✗	✗
compareTo()	✓	✗	✗
compareToIgnoreCase()	✓	✗	✗
concat()	✓	✗	✗
split()	✓	✗	✗
replaceAll()	✓	✗	✗
replaceFirst()	✓	✗	✗
matches()	✓	✗	✗
toLowerCase()	✓	✗	✗
toUpperCase()	✓	✗	✗
trim()	✓	✗	✗
strip()	✓	✗	✗
isEmpty()	✓	✗	✗
isBlank()	✓	✗	✗
repeat()	✓	✗	✗

17. The Big Picture

A simple way to explain the relationship among the three classes is shown below.



The Easiest Rule to Remember

String → operations mostly create a new String

```
String s = "Hello";  
s = s + " World";
```

StringBuilder → operations modify the same object

```
StringBuilder sb = new StringBuilder("Hello");  
sb.append(" World");
```

StringBuffer → like StringBuilder, but synchronized/thread-safe

```
StringBuffer sb = new StringBuffer("Hello");  
sb.append(" World");
```

Key Teaching Point

String, StringBuilder, and StringBuffer are three separate Java classes with overlapping APIs. They should not be treated as having exactly the same methods.

String is immutable. StringBuilder is mutable and is generally preferred for frequent string modifications in ordinary single-threaded programs. StringBuffer is mutable and synchronized/thread-safe.